

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Safety and Efficacy of Pembrolizumab (MK-3475) Coformulated With Berahyaluronidase Alfa (MK-3475A) in Participants With Relapsed or Refractory Classical Hodgkin Lymphoma (rrcHL) or Relapsed or Refractory Primary Mediastinal Large B-cell Lymphoma (rrPMBCL)(MK-3475A-F65) **Short Title/Acronym:** MK-3475A-F65 **Protocol Number:** MK-3475A-F65 **NCT Number:** NCT06504394 **Sponsor Name:** Merck **Investigational Product:** Pembrolizumab coformulated with berahyaluronidase alfa (hyaluronidase) (MK-3475A) **Phase of Development:** Phase 2 **Trial Status:** Recruiting **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the pharmacokinetics (PK) profile of pembrolizumab following subcutaneous (SC) injection of pembrolizumab coformulated with hyaluronidase, and to evaluate the objective response rate (ORR) of pembrolizumab (+) berahyaluronidase alfa SC in adult participants with Relapsed or Refractory Classical Hodgkin Lymphoma (rrcHL) or Relapsed or Refractory Primary Mediastinal Large B-cell Lymphoma (rrPMBCL). **Secondary Objectives:** To evaluate the duration of response (DOR), progression-free survival (PFS), overall survival (OS), antidrug antibodies (ADA), and the safety profile of the SC treatment. There is no formal hypothesis to be tested for this study.

2.2 Background & Rationale Pembrolizumab is an immunotherapy traditionally administered via intravenous (IV) infusion. Giving pembrolizumab under the skin as a subcutaneous (SC) injection coformulated with hyaluronidase can significantly reduce the administration time. This study aims to provide a more convenient treatment option that may increase access and save administration time for patients and healthcare providers, without compromising the treatment's efficacy and pharmacokinetics.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Targeted Enrollment: 60
participants **Number of Sites: Global Multicenter Estimated Study**
Start: 14-Oct-2024 Estimated Primary Completion: 08-Nov-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a life expectancy of > 3 months. 2. Histologically confirmed diagnosis of classical Hodgkin lymphoma (cHL) or primary mediastinal B-cell lymphoma (PMBCL). 3. Radiographically measurable cHL or PMBCL disease assessed by investigator as per Lugano classification. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 assessed within 7 days before or on the day of the first dose of study intervention. 5. Human immunodeficiency virus (HIV)-infected participants must have well controlled HIV on antiretroviral therapy (ART). 6. Participants who are hepatitis B surface antigen (HBsAg) positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy for at least 4 weeks, and have undetectable HBV viral load before enrollment. 7. Participants with a history of hepatitis C virus (HCV) infection are eligible if they have completed curative antiviral therapy at least 4 weeks before enrollment and HCV viral load is undetectable at screening.

4.2 Exclusion Criteria 1. Has clinically significant (i.e., active) cardiovascular disease. 2. Has pericardial effusion or clinically significant pleural effusion. 3. Has known additional malignancy that is progressing or has required active treatment within the past 2 years. 4. Diagnosis of immunodeficiency or is receiving systemic steroid therapy or any other form of immunosuppressive therapy within 7 days prior to the first dose of study intervention. 5. Received prior monoclonal antibody within 4 weeks prior to first dose of study intervention or has not recovered (i.e., $\leq$ Grade 1 or at baseline) from adverse events (AEs) due to agents administered more than 4 weeks earlier. 6. Received prior therapy with an anti-programmed cell death 1 protein (anti-PD-1), anti-programmed cell death ligand 1 (anti-PD-L1), or anti-programmed cell death ligand 2 (anti-PD-L2) agent, or with an agent directed to another stimulatory or coinhibitory T-cell receptor. 7. Received prior systemic anticancer therapy including investigational agents within 4 weeks before the first dose of study intervention. 8. Received prior radiotherapy within 2 weeks of start of study intervention, or has radiation-related toxicities requiring corticosteroids. 9. Received a live or live-attenuated vaccine within 30 days before first dose of study intervention. 10. Is receiving systemic antineoplastic chemotherapy, immunotherapy, or biological therapy not specified in this protocol. 11. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 12. Active autoimmune disease that has required systemic treatment in the past 2 years. 13. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or

has current pneumonitis/interstitial lung disease. 14. Active infection requiring systemic therapy except certain protocol-specified therapies. 15. Concurrent active hepatitis B and hepatitis C virus infection. 16. Has undergone solid organ transplant at any time, or prior allogeneic hematopoietic stem cell transplant (SCT) within the last 5 years.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab coformulated with hyaluronidase administered once every 6 weeks (Q6W). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Approximately 2 years (up to 18 treatments) unless disease progression or unacceptable toxicity occurs.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for the management of symptoms. **Prohibited:** Concurrent investigational agents or previous intravenous pembrolizumab/anti-PD-1 therapies, live vaccines within 30 days, unapproved systemic antineoplastic therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Day -28 to 0	Informed Consent, Medical History, Disease Assessment, Labs
2	Day 0	First SC Injection, Vital Signs, PK Blood Draws
3	Every 6 Weeks	SC Injection, Safety Labs, PK Monitoring, Radiographic Efficacy Assessment
4	Approx. Year 2	Final Vital Signs, Exit Interview, Disease Evaluation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Objective Response Rate (ORR)** per Lugano Classification Criteria as Assessed by Investigator. 2. **Maximum Concentration (C_{max})** of Pembrolizumab After Administration of SC Pembrolizumab Coformulated with Hyaluronidase. 3. **Lowest Plasma Concentration (C_{trough})** of Pembrolizumab After Administration of SC Pembrolizumab Coformulated with Hyaluronidase.

7.2 Secondary & Exploratory Endpoints 1. **Duration of Response (DOR)** per Lugano Classification Criteria as Assessed by Investigator. 2. **Number of Participants with Antidrug Antibodies (ADA)** Level of Pembrolizumab After Administration of SC Pembrolizumab Coformulated with Hyaluronidase. 3. **Maximum Concentration (C_{max})** of Pembrolizumab After Administration of SC Pembrolizumab Coformulated with Hyaluronidase at Steady State. 4. **Progression-Free Survival (PFS)** and **Overall Survival (OS)**.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite physical examinations, blood/urine laboratory tests, and patient reports during every 6-week cycle visit. **Serious Adverse Events (SAEs):** Routine reporting timeline, typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal sponsor-designated committee to oversee safety and trial conduct.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy outcomes; Safety Set for all adverse events; PK analysis set for pharmacokinetic parameters. **Sample Size Justification:** Target enrollment of 60 is considered adequately powered to evaluate the pharmacokinetic profile and clinical efficacy response rate in a Phase 2 setting. **Handling of Missing Data:** Standard survival analysis algorithms (e.g., Kaplan-Meier) for time-to-event endpoints and imputation rules as specified by the SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure data integrity, protocol adherence, and patient safety. **Audit Trail:** Details on eCRF locking, source data verification, and data clarification forms will be strictly maintained by the sponsor.

11. Study Identifiers & Governance

Primary ID: MK-3475A-F65 **Registry Number:** NCT06504394 **Posting ID:** 403403033384798878 **Sponsor/Lead Organization:** Merck **Company:** Merck **Study Title:** MK-3475A-F65 **Official Regulatory Title:** A Phase 2 Study to Evaluate the Pharmacokinetics, Efficacy, and Safety of Subcutaneous Pembrolizumab Coformulated With Hyaluronidase (MK-3475A) in Participants With Relapsed or Refractory Classical Hodgkin Lymphoma (rrCHL) or Relapsed or Refractory Primary Mediastinal Large B-cell Lymphoma (rrPMBCL)

12. Clinical Framework (Lymphoma Specific)

Disease Areas: Classical Hodgkin Lymphoma Recurrent, Classical Hodgkin Lymphoma Refractory, Primary Mediastinal Large B-cell Lymphoma Recurrent, Primary Mediastinal Large B-cell Lymphoma Refractory **Diagnosis Threshold:** Histologically confirmed cHL or

PMBCL, with radiographically measurable disease **Medical Methodology:** Anti-PD-1 Immunotherapy Coformulation **Optimization Target:** Pharmacokinetic profile (Cmax, Ctrough) and Objective Response Rate (ORR)

13. Study Procedures & Methodology

Management Pathway: Standard oncology pathway for relapsed/refractory lymphoma with transition to SC immunotherapy **Education Protocol (HCP):** Site training on subcutaneous administration and AE management **Education Protocol (Patient):** Education on SC formulation over IV infusion, expected treatment schedule (Q6W), and reporting AEs **Quality Audit Mechanism:** Routine clinical site monitoring, PK sample tracking, and efficacy evaluations

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 3 years per patient (about 2 years treatment + follow-up) **Onsite Visit Cadence:** Every 6 weeks (Q6W) corresponding with treatment administration **Remote Monitoring Frequency:** As needed between cycles for toxicity check-ins **Checklist Review Interval:** Prior to every Q6W dose

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]